

BORIS LI

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✉ boris.li@uwaterloo.ca

Education

University of Waterloo, Waterloo, ON
PhD Pure Mathematics

Sep 2026 –

- Supervisors: Drs. Jason Bell and Ben Webster

University of Waterloo, Waterloo, ON
MMath Pure Mathematics

Sep 2025 – Aug 2026

- Supervisor: Dr. Jason Bell

University of British Columbia, Vancouver, BC
BSc Combined Honours Physics & Mathematics *with Distinction*

Sep 2020 – Apr 2025

- Honours thesis supervisor: Dr. Ian Frigaard
- 2-time Dean's Scholar (90% or better average)
- 2-time Dean's List (80% or better average)
- Science One, class of 2021

Publications

Jason Bell and **Boris Li**. “Filtered deformations of three-variable polynomial algebras”. Preprint, (2026)
arXiv: [2609.06710](#).

Boris Li. “Filtered Deformations of Trivariate Polynomial Rings over a Field”. MMath research paper, (2026).

Boris Li. “Stability of Two-fluid Concentration Gradients”. BSc thesis, (2024).

A. Abghari, E. F. Bunn, L. T. Hergt, **B. Li**, D. Scott, R. M. Sullivan, and D. Wei. “Reassessment of the dipole in the distribution of quasars on the sky”. Journal of Cosmology and Astroparticle Physics, 11(2024)067.
arXiv: [2405.09762](#).

- Contribution: Initial data analysis

Teaching

University of Waterloo, Waterloo, ON
Department of Pure Mathematics
Graduate Teaching Assistant

Sep 2025 –

- MATH 116 (Calculus 1 for Engineering): 1 term
- MATH 117 (Calculus 1 for Engineering): 1 term
- MATH 136 (Linear Algebra 1 for Honours Math): 1 term
- MATH 235 (Linear Algebra 2 for Honours Math): 1 term

- PMATH 336 (Intro Group Theory and Applications): 1 term
- PMATH 340 (Elementary Number Theory): 1 term
- PMATH 347 (Groups and Rings): 1 term
- PMATH 450/650 (Measure Theory): 1 term

University of British Columbia, Vancouver, BC
Science One Program & Department of Computer Science
 Undergraduate Teaching Assistant

Sep 2021 – Apr 2024

- SCIE 001 (Science One Program): physics and mathematics portions, 3 years
- CPSC 110 (Computation, Programs, and Programming): 6 terms

Employment

University of Waterloo, Waterloo, ON
Faculty of Mathematics
 Women in Mathematics Program Coordinator

May 2026 –

- Maintained the WiM Directed Reading Program website and application portal
- Implemented frontend user features using Angular
- Maintained the backend server and database, which uses Nodejs with Express routing, and Sequelize to interact with a sqlite3 database

University of British Columbia, Vancouver, BC
Departments of Mathematics and Mechanical Engineering
 Undergraduate Research Assistant

May – Aug 2023

- Supervised by Dr. Ian Frigaard
- Studied the fluid mechanics behind flow between a heavier Herschel-Bulkley fluid on top of a lighter Newtonian fluid in a vertical pipe
- Utilized both experimental techniques, and simulations through ParaFOAM

University of British Columbia, Vancouver, BC
Department of Physics and Astronomy
 Undergraduate Research Assistant

May – Aug 2022

- Supervised by Dr. Douglas Scott
- Studied the dipole anisotropy of quasar distributions from WISE data
- Utilized astronomy-related Python packages, such as Astropy, Healpy, and PolSpice

University of British Columbia, Vancouver, BC
Department of Mathematics
 Undergraduate Academic Assistant

May – Aug 2021

- Wrote and designed questions for PrairieLearn, an online homework and testing platform, based on the CLP textbooks¹
- Developed a system for units-aware problems in PrairieLearn, which has now been merged into the main repository²³

¹<https://personal.math.ubc.ca/~CLP/>

²<https://github.com/PrairieLearn/PrairieLearn/pull/4801>

³<https://github.com/PrairieLearn/PrairieLearn/pull/6790>

Scholarships & Awards

Ontario Graduate Scholarship (CAD\$15000) Ontario Student Assistant Program	2026
President's Graduate Scholarship (CAD\$10000) Graduate Studies and Postdoctoral Affairs, University of Waterloo	2026
Undergraduate Student Research Award (CAD\$12000 over 2 awards) National Sciences and Engineering Research Council of Canada	2022 – 2023
Stanley M Grant Scholarship in Mathematics (CAD\$4350 over 2 awards) Department of Mathematics, University of British Columbia	2022 – 2023
Charles and Jane Banks Scholarship (CAD\$490 over 2 awards) Faculty of Science, University of British Columbia	2021 – 2022
Trek Excellence Scholarship (CAD\$1500) University of British Columbia	2022